Chapter 1 — Introduction

1.1 Background and Motivation

Air pollution is one of the most pressing environmental challenges of the 21st century. Rapid urbanization, increasing vehicular traffic, industrial emissions, and agricultural burning have collectively degraded air quality across cities and towns worldwide. In India, cities such as Gandhinagar and Ahmedabad regularly record Air Quality Index (AQI) values that exceed safe limits, posing serious public health risks including respiratory illness, cardiovascular disease, and premature mortality. Conventional air quality monitoring infrastructure is expensive, bulky, and limited in geographic coverage. This project addresses that gap by designing a compact, low-cost, multi-sensor air pollution monitoring system using readily available IoT components that can be assembled and deployed by students and hobbyists.

1.2 Problem Statement

Most individuals, particularly in smaller cities and residential areas, have no access to real-time, localized air quality data. Existing monitoring stations are few in number, difficult to access, and do not provide a granular picture of pollution at street level. This project proposes a self-contained hardware solution that measures multiple air quality parameters simultaneously and provides immediate visual and audio alerts when dangerous thresholds are crossed.

1.3 Objectives

- Measure concentration of multiple gases including CO2, ammonia, benzene, LPG, smoke, and carbon monoxide.
- Detect airborne particulate matter (dust) using an optical sensor.
- Measure ambient temperature and relative humidity.
- Calculate the overall Air Quality Index (AQI) from all sensor inputs.
- Display real-time readings on a 16x2 LCD in rotating pages.
- Provide color-coded visual alerts using an RGB LED.
- Trigger an audible buzzer alarm when AQI reaches hazardous levels.
- Power the high-current MQ sensors safely using an external LM7805 regulated supply.

Chapter 2 — System Overview

2.1 Architecture

The system follows a three-layer architecture: the sensing layer collects raw environmental data, the processing layer (Arduino Uno) calculates AQI and determines alert status, and the output layer communicates results through the LCD, RGB LED, and buzzer.

Sensing Layer → Arduino Uno (Processing) → LCD + RGB LED + Buzzer

2.2 Data Flow

Each sensor produces an analog output voltage proportional to the concentration of its target pollutant. The Arduino reads these voltages through its ADC pins at regular intervals. Raw ADC values (0–1023) are mapped to an AQI scale (0–500). The Sharp dust sensor requires a precisely timed LED pulse sequence to produce a valid reading. The DHT22 communicates via a proprietary single-wire digital protocol. All readings are processed together to compute a final AQI, which drives the output devices. The LCD rotates through three display pages every three seconds to show all measured parameters without requiring additional hardware.

Chapter 3 — Hardware Components

3.1 Arduino Uno (ATmega328P)

The Arduino Uno is the central processing unit of the system. It is based on the 8-bit ATmega328P microcontroller running at 16 MHz. It provides 14 digital I/O pins (6 of which support PWM output), 6 analog input pins with 10-bit ADC resolution (0-1023), and interfaces for UART, SPI, and 12C communication. It is programmed using the Arduino IDE with C/C++ and powered via USB from a laptop or charger. PWM capability on pins D3, D5, D6, D9, D10, and D11 is used to mix RGB colours on the LED.

3.2 MQ-135 Air Quality Sensor

The MQ-135 detects a broad spectrum of harmful gases including ammonia (NH3), benzene (C6H6), carbon dioxide (CO2), alcohol vapour, and smoke. Its sensing element is tin dioxide (SnO2), whose electrical conductivity increases when target gases are present. It operates at 5V and draws approximately 150mA due to its internal heater. The analog output voltage is read on Arduino pin AO. It is powered from the external LM7805 supply rather than the Arduino's 5V pin due to its high current draw.

3.3 MQ-2 Smoke and LPG Sensor

The MQ-2 is sensitive to combustible gases including LPG, propane, hydrogen, and smoke. It uses the same SnO2, sensing principle as MQ-135 and also draws approximately 150mA. It is connected to analog pin A1 and powered from the external supply. The MQ-2 is particularly useful for fire hazard detection and gas leak alarms in indoor environments.

3.4 MQ-7 Carbon Monoxide Sensor

The MQ-7 is specifically calibrated to detect carbon monoxide (CO), a colorless and odorless gas produced by incomplete combustion. It is one of the most dangerous household pollutants, causing headaches, dizziness, and death at high concentrations. The MQ-7 operates at 5V with a current draw of approximately 150mA and produces an analog output connected to pin A2, powered from the external supply.

3.5 Sharp GP2Y1010AU0F Optical Dust Sensor

Unlike the MQ series which detects gases, the Sharp GP2Y1010AU0F detects solid and liquid airborne particles (dust, smoke particles, pollen) using an infrared LED and a phototransistor. It outputs an analog voltage proportional to dust density in mg/m3. The internal LED must be pulsed — activated for exactly 280 microseconds before reading the output voltage, then deactivated. This timing is handled precisely in the firmware using delayMicroseconds(). The sensor requires a 150 ohm resistor on its V-LED pin and a 220 microfarad electrolytic capacitor for LED current stabilization. The analog output connects to pin A3 and the LED control connects to pin D9.

3.6 DHT22 Temperature and Humidity Sensor

The DHT22 (AM2302) measures ambient temperature from -40 to +80 degrees Celsius with 0.5 degree accuracy, and relative humidity from 0 to 100% with 2% accuracy. It communicates via a single-wire digital protocol and is read using the Adafruit DHT library. A 10k ohm pull-up resistor is required between the data pin and VCC. It connects to digital pin D8 and operates at 5V drawing only 2.5mA, so it is powered directly from the Arduino's 5V pin. Temperature and humidity values are displayed on the LCD's third page.

3.7 16x2 LCD Display

The 16x2 character LCD displays two rows of up to 16 characters each. It is connected in 4-bit mode using pins RS=D2, EN=D3, D4=D4, D5=D5, D6=D6, D7=D7. Contrast is adjusted using a 10k potentiometer connected to the VO pin. The LCD backlight (pins A and K) is connected directly to 5V and GND respectively. Three rotating display pages are implemented in firmware: Page 1 shows AQI and status, Page 2 shows gas and dust raw values, and Page 3 shows temperature and humidity.

3.8 RGB LED and Buzzer

A common-cathode RGB LED combines three LEDs (red, green, blue) in a single package. By mixing PWM signals on pins D11 (Red), D12 (Green), and D13 (Blue), the firmware produces five distinct colors corresponding to five AQI levels — green for good, yellow for moderate, orange for unhealthy, red for very unhealthy, and purple for hazardous. Each LED leg uses a 220 ohm currentlimiting resistor. The buzzer connects to pin D10 with its negative terminal to GND and activates only when AQI exceeds 300 (hazardous).

3.9 LM7805 Voltage Regulator Power Supply

The three MQ sensors together draw approximately 450mA, which exceeds the Arduino Uno's 5V regulator limit of 500mA when combined with other components. A dedicated LM7805 linear voltage regulator circuit is used to power the MQ sensors from a 9V DC adapter. The circuit uses a 47 microfarad 16V electrolytic capacitor on the input and a 1 microfarad 16V electrolytic capacitor on the output for stability. The GND of this external circuit is connected to the Arduino GND pin to establish a shared reference, allowing sensor analog signals to be correctly interpreted by the Arduino ADC.